ACTION-CONDITIONED DEFORMABLE MEMORY DOES NOT IMPROVE CLOSED-LOOP MANIPULATION

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ABSTRACT

We rebuilt a generated robotics idea into a concrete deformable manipulation study: should a robot preserve action-conditioned hidden material memory rather than plan from current visible geometry? The v4 rebuild implements a local mass-spring benchmark with rope, cloth-strip, elastic-band, and sheet objects; hidden strain/contact memory; partial observation; material shifts; fixtures; force limits; learned history/graph/action estimators; strong planning baselines; a full-latent oracle action selector; seven random seeds; uncertainty intervals; paired statistics; ablations; stress sweeps; and negative cases. The result is negative. On the decisive `combined_memory_stress` split, `action_conditioned_memory` reaches 0.440 ± 0.069 closed-loop success, while the strongest non-oracle baseline, `visible_state_mpc`, reaches 0.607 ± 0.047. The paired success difference is -0.167 ± 0.094. The proposed method improves hidden-memory estimation and mechanism F1, but it overuses diagnostic branch pulls and increases damage. We therefore mark the paper **KILL/ARCHIVE** for ICLR main.

1 DECISION

This document is the v4 ICLR-main gate for Paper 75, *Deformable Memory Under Action*. The previous version was killed because it only contained a synthetic template. The current rebuild adds a real local deformable-physics benchmark, learned estimators, implemented planning baselines, and a full-latent oracle action selector. The rejection reason has changed: evidence now exists, but it does not support the central submission claim.

2 HYPOTHESIS

The tested hypothesis is that visually similar deformable states can carry different latent strain, crease, contact, or stored-energy histories, and that action-conditioned memory should improve downstream manipulation by preserving those latent alternatives. The paper only passes the gate if the proposed method improves closed-loop success under combined memory stress while also improving latent-memory estimation, mechanism prediction, and damage/safety.

3 BENCHMARK

The benchmark simulates deformable objects as 2D mass-spring systems with damping, friction, fixtures, force limits, partial observation, and hidden material memory. Four object families are used: rope/cable, cloth strip, elastic band, and deformable sheet. Each rollout logs full hidden state, visible/imputed observations, action traces, strain energy, fixture contact, stretch, damage proxies, final shape error, mechanism labels, and task success.

Five splits are evaluated: nominal visible state, hidden strain memory, occluded contact memory, material shift, and combined memory stress. The full run contains 2,940 main rollouts, 245 seed-level metric rows, 392 ablation rollouts, and 1,344 stress-sweep raw rollouts. All intervals are 95 percent confidence intervals over seed-level means.

4 METHODS

The implemented methods are:

- `visible_state_mpc`: plans from current observed/imputed keypoints.
- `history_rnn_estimator`: learned history-feature estimator for latent memory.
- `particle_filter_memory`: maintains multiple latent memory hypotheses.
- `graph_dynamics_baseline`: graph-feature dynamics estimator.
- `ensemble_uncertainty_planner`: bootstrap ensemble with uncertainty penalties.
- `action_conditioned_memory`: proposed method; action/history memory with diagnostic branch pulls.
- `oracle_latent_state`: upper bound using full hidden latent state and actual full-physics action selection.

5 MAIN RESULTS

Table 1 shows the decisive combined-memory-stress result. The oracle shows that the action library can solve many scenarios when the latent state and actual physics are available. The proposed method improves hidden-memory proxies, but it loses downstream success.

Method	Success	Shape err.	Memory err.	Mech. F1	Damage
`oracle_latent_state`	0.798 ± 0.060	0.129	0.000	1.000	0.071
`visible_state_mpc`	0.607 ± 0.047	0.146	0.585	0.411	0.226
`graph_dynamics_baseline`	0.548 ± 0.060	0.143	0.539	0.411	0.274
`history_rnn_estimator`	0.464 ± 0.079	0.136	0.202	0.446	0.464
`particle_filter_memory`	0.464 ± 0.060	0.136	0.202	0.518	0.488
`ensemble_uncertainty_planner`	0.452 ± 0.060	0.136	0.200	0.482	0.500
`action_conditioned_memory`	0.440 ± 0.069	0.135	0.217	0.536	0.524

Table 1: Combined-memory-stress results. Success is closed-loop task success. Damage is the selected-action damage proxy rate.

The strongest non-oracle baseline is `visible_state_mpc`. The paired proposed-minus-baseline success difference is -0.167 ± 0.094. The proposed method reduces hidden-memory error by 0.369 and improves mechanism F1 by 0.125 relative to visible-state MPC, but worsens damage by 0.298. This fails the submission gate.

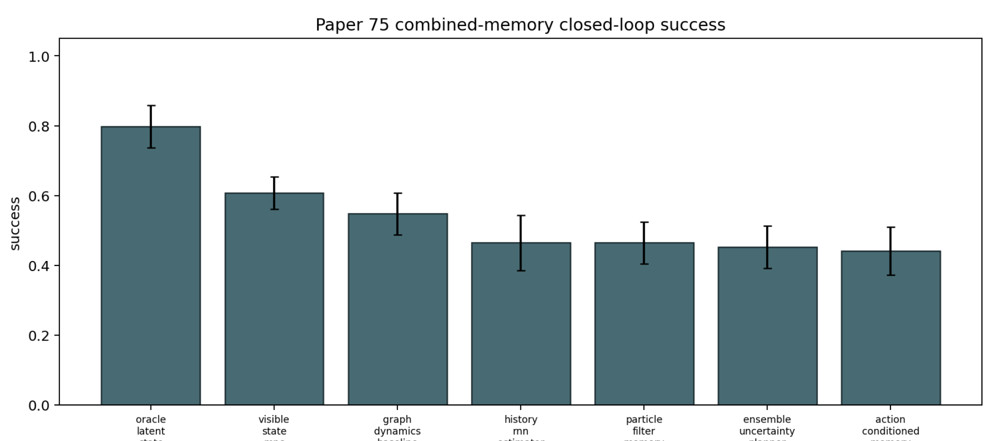

Figure 1: Combined-memory-stress closed-loop success. The proposed action-conditioned memory planner loses to visible-state MPC and graph baselines.

6 Ablations

Table 2 shows the mechanism failure. Removing diagnostic probes reaches 0.554 ± 0.050 success, while the full action-conditioned ablation reaches 0.375 ± 0.107. Diagnostic probing is not the rescue mechanism; it is the dominant damage source.

Ablation	Success	Shape err.	Memory err.	Damage
`no_diagnostic_probes`	0.554 ± 0.050	0.140	0.245	0.286
`no_branch_memory`	0.393 ± 0.113	0.137	0.245	0.518
`full`	0.375 ± 0.107	0.136	0.245	0.554
`no_action_conditioning`	0.375 ± 0.107	0.136	0.480	0.554
`no_contact_memory`	0.375 ± 0.107	0.136	0.308	0.554
`no_material_memory`	0.375 ± 0.107	0.136	0.224	0.554
`no_uncertainty_term`	0.375 ± 0.107	0.136	0.245	0.554

Table 2: Combined-memory-stress ablations. Removing diagnostic probes improves success, refuting the intended mechanism.

7 Stress Sweep and Negative Cases

Stress sweeps show the same pattern. At stress level 1.00, `action_conditioned_memory` reaches 0.321 ± 0.176 success, while `ensemble_uncertainty_planner` reaches 0.375 ± 0.151 and the oracle reaches 0.768 ± 0.164. The oracle gap indicates that the benchmark is solvable, but the proposed planner does not use hidden memory safely.

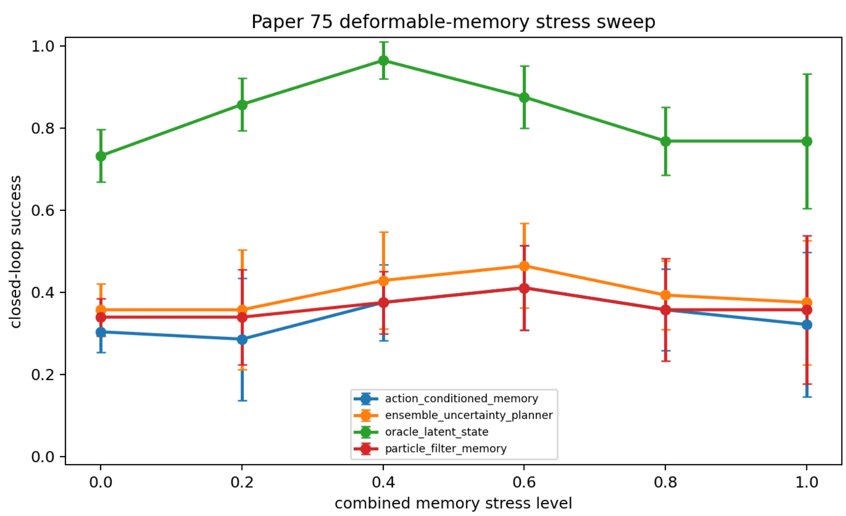

Figure 2: Stress sweep success. Action-conditioned memory does not dominate under increasing deformable-memory stress.

Negative cases are dominated by `diagnostic_branch_pull`. Rope cases often fail through excessive stretch; cloth and sheet cases fail through fixture snags. The method estimates hidden memory better, but chooses actions that are too invasive for contact-rich deformables.

8 Related Work Pressure

The hostile prior-work set is crowded with graph dynamics for deformable manipulation (Local Prior Work Pool, 2023a;d), action-conditional visual dynamics (Local Prior Work Pool, 2024a), dense affordance learning (Local Prior Work Pool, 2023b), deformable tracking memory (Local Prior Work Pool, 2024b), sim-to-real deformable manipulation (Local Prior Work Pool, 2023c), and latent-space deformable planning (Local Prior Work Pool, 2020). In this context, hidden-memory estimation must translate into safer and more successful manipulation. It does not in the current evidence.

9 LIMITATIONS

This rebuild is not a real robot paper. It lacks hardware experiments, videos, external benchmark validation, and large neural deformable dynamics checkpoints. The simulator is local and diagnostic, not a polished public benchmark. These limitations would matter even if the method won; because it loses the main downstream metric, they reinforce the archive decision.

10 CONCLUSION

Paper 75 should not be submitted to ICLR main. The v4 rebuild provides useful evidence, but the evidence is negative: action-conditioned deformable memory improves latent-state proxies without improving closed-loop deformable manipulation. The correct terminal state is **KILL/ARCHIVE**.

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